Computational Validation of PET Degradation Activity

Engineered IsPETase Variant — Step-by-Step Reference Guide

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This guide covers three levels of computational validation for whether the engineered IsPETase variant retains PET-degrading activity after the 8 mutations. Each level builds on the previous. Complete Level 1 first, then proceed to Levels 2 and 3 as time permits. All tools referenced are free.

Overview of all three levels

Level	What it tests	Difficulty	Time	Impact
1	Active site geometry preserved after mutations	Easy	1-2 hours	**Strong**
2	PET substrate binds correctly in engineered variant	Moderate	1 full day	**Very strong**
3	PET stays stably bound during MD simulation at pH 7.0	Hard	Several days	**Publication-level**

Level 1 — Active Site Geometry Verification via AlphaFold2

What this level tests

Whether the catalytic triad geometry — the precise spatial arrangement of Ser160, Asp206, and His237 — is preserved in your engineered variant after the 8 mutations. If the three residues are still in the correct positions and orientations relative to each other, the enzyme's cutting mechanism is intact and PET hydrolysis can proceed.

The biology: IsPETase cuts PET by a three-step relay. Ser160 attacks the carbonyl carbon of the PET ester bond. His237 acts as a base that activates Ser160. Asp206 stabilizes His237. If any of these three residues moves even 1-2 Angstroms from its correct position, the relay breaks and the enzyme loses activity entirely.

Tool needed: AlphaFold2 (free, no installation required)

Option A:	`https://alphafold.ebi.ac.uk`
Option B:	`https://colab.research.google.com/github/deepmind/alphafold/blob/main/notebooks/AlphaFold.ipynb`
Viewer:	`https://www.cgl.ucsf.edu/chimerax/download.html (UCSF ChimeraX -- free)`

Step-by-step instructions

Step 1 — Get your engineered sequence

Open your file: results/step3_mutations/engineered_IsPETase_variant.fasta

Copy the 290 amino acid sequence (everything after the > header line).

Step 2 — Run AlphaFold2

Go to https://alphafold.ebi.ac.uk and click 'Predict structure'.

Paste your 290 aa engineered sequence into the input box.

Click Submit. Wait 10-20 minutes for the prediction to complete.

Download the predicted PDB file when done.

Step 3 — Also download the wild-type structure

Go to https://www.rcsb.org and search '5XJH'.

Click Download Files > PDB Format.

Save as 5XJH_wildtype.pdb.

Step 4 — Open both structures in ChimeraX

Install UCSF ChimeraX from the link above.

Open ChimeraX. Use File > Open to load both PDB files.

Both structures will appear in the 3D viewer simultaneously.

Step 5 — Align the two structures

In the ChimeraX command line at the bottom, type:

```
matchmaker #1 to #2
```

This superimposes the engineered structure onto the wild-type using all backbone atoms.

Step 6 — Measure catalytic triad distances

In the ChimeraX command line, type the following commands one at a time.

Replace #1 with whichever model number is your ENGINEERED structure:

```
distance #1:160@OG #1:237@NE2
distance #1:237@NE2 #1:206@OD1
distance #1:160@OG #1:206@OD1
```

Then repeat for model #2 (wild-type 5XJH) to get comparison values.

Step 7 — Record your measurements

Write down all 6 distances (3 for engineered, 3 for wild-type).

Compare each pair. If engineered distances are within 0.5 Angstroms of

wild-type values at all three contacts, the catalytic machinery is intact.

What to measure — target values

Contact	Target distance	Significance
Ser160 OG to His237 NE2	`<= 3.2 Angstroms`	Ser160 activation relay
His237 NE2 to Asp206 OD1	`<= 3.2 Angstroms`	His237 stabilization
Ser160 OG to substrate carbon	`<= 3.5 Angstroms`	Nucleophilic attack geometry

> **What to write in your paper: AlphaFold2 structure prediction of the engineered variant showed catalytic triad geometry within X Angstroms of the wild-type crystal structure across all three catalytic contacts (Ser160-His237: X A, His237-Asp206: X A), supporting the prediction of retained PET-hydrolyzing activity.**

Level 2 — Molecular Docking of PET Substrate (AutoDock Vina)

What this level tests

Whether BHET (the PET monomer used in all IsPETase activity assays) can physically fit into the substrate binding cleft of your engineered variant with a productive binding pose — specifically whether your R280E mutation in subsite IIc helps or hurts PET accommodation compared to wild-type R280 and the published R280A mutation.

> **Why R280E specifically: Joo et al. (2018) showed R280A increased PET degradation by 22-32% by relieving steric obstruction in subsite IIc. Your R280E introduces Glutamate instead of Alanine at the same position. Docking will show whether Glutamate performs comparably, better, or worse than Alanine for substrate accommodation -- a genuinely novel comparison not in any published paper.**

Tools needed

AutoDock Vina:	`https://vina.scripps.edu/downloads/`
AutoDockTools:	`https://ccsb.scripps.edu/mgltools/downloads/`
BHET structure:	`https://pubchem.ncbi.nlm.nih.gov/compound/Bis_2-hydroxyethyl_terephthalate`
OpenBabel:	`https://openbabel.org/wiki/Get_Open_Babel (for file conversion)`

Step-by-step instructions

Step 1 — Download and install AutoDock Vina

Go to https://vina.scripps.edu/downloads/ and download Vina for Windows.

Also download AutoDockTools (MGLTools) from the link above.

Install both. AutoDockTools provides the graphical interface for setup.

Step 2 — Download BHET substrate structure

Go to PubChem: https://pubchem.ncbi.nlm.nih.gov/compound/68442

Click Download > SDF > 3D Conformer.

Save as BHET.sdf

Step 3 — Prepare your protein structures

You need two protein files: your engineered AlphaFold structure (from Level 1)

and the wild-type 5XJH structure.

Open AutoDockTools. Load each protein PDB file.

Use Edit > Hydrogens > Add (polar only) to add hydrogens.

Save each as .pdbqt format using File > Save > Write PDBQT.

Step 4 — Prepare the BHET ligand

Open AutoDockTools. Load BHET.sdf.

Use Ligand > Input > Open to load it.

Use Ligand > Torsion Tree > Detect Root to set rotatable bonds.

Save as BHET.pdbqt using Ligand > Output > Save as PDBQT.

Step 5 — Define the docking search box

In AutoDockTools, load your protein structure.

Use Grid > Grid Box to define the search region.

Center the box on Ser160. Use these approximate settings:

Center X/Y/Z: coordinates of Ser160 CA atom from your structure

Box size: 20 x 20 x 20 Angstroms

This ensures the entire substrate binding cleft is included.

Note the center coordinates -- you need them for the config file.

Step 6 — Create the Vina configuration file

Create a plain text file called vina_config.txt with this content:

```
receptor = engineered_IsPETase.pdbqt
ligand = BHET.pdbqt
center_x = [Ser160 X coordinate]
center_y = [Ser160 Y coordinate]
center_z = [Ser160 Z coordinate]
size_x = 20
size_y = 20
size_z = 20
exhaustiveness = 8
num_modes = 9
energy_range = 3
```

Step 7 — Run AutoDock Vina

Open Command Prompt. Navigate to your AutoDock Vina folder.

Run: vina --config vina_config.txt --out engineered_docked.pdbqt --log engineered_log.txt

Wait for completion (5-15 minutes).

Repeat with wildtype_IsPETase.pdbqt as the receptor to get comparison values.

Step 8 — Read the results

Open engineered_log.txt and wildtype_log.txt.

The first line of results shows the best binding pose energy in kcal/mol.

More negative = tighter binding.

Load the docked output .pdbqt files in ChimeraX alongside your protein

to visualize how BHET sits in the binding pocket.

Step 9 — Measure the key distance

In ChimeraX, with your docked complex loaded:

```
distance [BHET carbonyl carbon] [Ser160 OG]
```

This distance should be 3.5 Angstroms or less for a productive binding pose.

Record this for both engineered and wild-type.

What to measure — binding results

Measurement	Target	Significance
Binding energy (engineered)	`Compare to WT`	More negative = tighter binding
Binding energy (wild-type 5XJH)	`Reference value`	Your baseline for comparison
BHET carbonyl to Ser160 OG	`<= 3.5 Angstroms`	Productive binding pose confirmed
Substrate pose in subsite I	`Between Tyr87 and Trp185`	Pi-pi stacking interaction
R280E effect on subsite IIc	`Compare to R280 pose`	Novel R280E vs R280A finding

> **What to write in your paper: Molecular docking of BHET into the AlphaFold2-predicted engineered structure yielded a binding energy of X kcal/mol compared to X kcal/mol for wild-type, with the substrate adopting a productive pose positioning the scissile ester bond X Angstroms from Ser160 OG. The R280E substitution in subsite IIc produced [comparable/improved/reduced] substrate accommodation relative to the published R280A variant.**

Level 3 — MD Simulation with PET Substrate Bound (OpenMM)

What this level tests

Whether BHET remains stably bound in a productive orientation inside the active site of your engineered variant during a molecular dynamics simulation at pH 7.0 — the conditions in the small intestine where the enzyme would actually perform PET degradation after surviving gastric transit. This is the most rigorous computational validation possible short of wet lab experiments.

> **Why pH 7.0 and not pH 1.2: IsPETase is designed to degrade PET at neutral pH. The stomach at pH 1.2 is only the transit environment. Once the enzyme survives and reaches the small intestine (pH 6-7), it should perform degradation there. This simulation tests whether the enzyme-substrate complex is stable at the pH where activity actually occurs.**

Tools needed (install in your venv)

```
pip install openmm pdbfixer mdtraj
# Also requires GAFF2 force field for BHET ligand parameters
# Use ACPYPE on Google Colab (free) to generate BHET force field files
```

ACPYPE Colab:	`https://colab.research.google.com/github/alanwilter/acpype/blob/master/acpype_colab.ipynb`

Step-by-step instructions

Step 1 — Generate BHET force field parameters

Go to the ACPYPE Google Colab notebook (link above).

Upload your BHET.sdf file.

Run the notebook -- it generates GAFF2 force field parameters for BHET.

Download the output .frcmod and .mol2 files.

These files tell OpenMM how BHET atoms interact with each other and the protein.

Step 2 — Prepare the starting complex structure

Take the best docking pose from Level 2 (the .pdbqt output file).

Convert it to PDB format using OpenBabel:

```
obabel engineered_docked.pdbqt -O engineered_complex.pdb
```

This gives you a PDB file with both the protein and BHET in place.

Step 3 — Prepare the system with PDBFixer at pH 7.0

Run this Python script (save as prep_complex.py):

```
from pdbfixer import PDBFixer
from openmm.app import PDBFile
fixer = PDBFixer(filename='engineered_complex.pdb')
fixer.findMissingResidues()
fixer.findMissingAtoms()
fixer.addMissingAtoms()
fixer.addMissingHydrogens(7.0) # pH 7.0
PDBFile.writeFile(fixer.topology, fixer.positions,
```

open('complex_pH7_fixed.pdb', 'w'))

Step 4 — Run the MD simulation

The simulation setup follows the same OpenMM pattern as Phase 6 but includes

the BHET ligand in the simulation box. Key parameters:

Force field: AMBER14 protein + GAFF2 for BHET + TIP3P water

Temperature: 300 K (37 degrees C body temperature)

pH: 7.0 protonation states (from PDBFixer step above)

Simulation length: minimum 1 nanosecond for meaningful results

Save coordinates every 10 picoseconds for trajectory analysis.

Step 5 — Analyze substrate stability with MDTraj

After the simulation, analyze the trajectory:

```
import mdtraj as md
traj = md.load('simulation.dcd', top='complex_pH7_fixed.pdb')
```

Measure BHET RMSD relative to starting pose

```
bhet_rmsd = md.rmsd(traj, traj, 0, atom_indices=bhet_atom_indices)
```

Measure Ser160 OG to BHET carbonyl distance over time

```
dist = md.compute_distances(traj, [[ser160_og_idx, bhet_carbon_idx]])
```

Plot both over simulation time

Step 6 — Compare engineered vs wild-type

Repeat steps 2-5 with the wild-type 5XJH structure and BHET docked into it.

Compare:

```
- Mean BHET RMSD over the trajectory (lower = more stable binding)
```

- Fraction of simulation time where Ser160-BHET distance < 3.5 A

```
(higher = more productive binding time)
```

- Stability of subsite I pi-pi stacking (Tyr87/Trp185 to BHET benzene ring)

What to measure — MD results

Measurement	Target	Significance
BHET RMSD (engineered)	`Compare to WT`	Lower = more stable substrate binding
BHET RMSD (wild-type)	`Reference`	Baseline comparison
Ser160-BHET distance	`< 3.5 A > 50% of simulation`	Productive binding maintained
Tyr87/Trp185 pi-pi distance	`3.3-4.0 Angstroms`	Subsite I interaction intact
BHET displacement from pose	`< 2.0 A RMSD`	Substrate not ejected from cleft

What to write in your paper: MD simulation of the engineered IsPETase-BHET complex at pH 7.0 over X nanoseconds showed mean substrate RMSD of X Angstroms compared to X Angstroms for wild-type, with the scissile ester bond maintained within productive distance of Ser160 for X% of the simulation time. These results support the prediction that the engineered variant retains full PET-hydrolyzing activity under small intestinal pH conditions following gastric transit.

Combined Paper Statement — After Completing All Three Levels

Once you complete all three levels, this is the paragraph you can write in your results or discussion section:

> **Computational assessment of the engineered variant's substrate binding capacity was conducted at three levels. First, AlphaFold2 structure prediction confirmed catalytic triad geometry within X Angstroms of the wild-type crystal structure across all three catalytic contacts. Second, molecular docking of BHET into the predicted engineered structure yielded a binding energy of X kcal/mol compared to X kcal/mol for wild-type, with the substrate adopting a productive pose positioning the scissile ester bond X Angstroms from Ser160 OG. Third, MD simulation of the enzyme-substrate complex at pH 7.0 over X nanoseconds showed mean BHET RMSD of X Angstroms relative to the starting pose, with productive Ser160-substrate geometry maintained for X% of simulation time. Together these analyses provide computational evidence that the eight-mutation engineered variant retains the structural and dynamic prerequisites for PET hydrolysis under small intestinal conditions following gastric transit.**

Quick reference — all URLs

Tool	URL
AlphaFold2 EBI	`https://alphafold.ebi.ac.uk`
AlphaFold2 Colab	`https://colab.research.google.com/github/deepmind/alphafold/blob/main/notebooks/AlphaFold.ipynb`
ChimeraX download	`https://www.cgl.ucsf.edu/chimerax/download.html`
AutoDock Vina	`https://vina.scripps.edu/downloads/`
AutoDockTools	`https://ccsb.scripps.edu/mgltools/downloads/`
BHET on PubChem	`https://pubchem.ncbi.nlm.nih.gov/compound/68442`
OpenBabel	`https://openbabel.org/wiki/Get_Open_Babel`
ACPYPE Colab	`https://colab.research.google.com/github/alanwilter/acpype/blob/master/acpype_colab.ipynb`